

An ideal non-extended formulation for leaky ReLU

February 1, 2023

1 Objective

For a leaky ReLU unit

$$y = \max(c(w^T x + b), w^T x + b) \quad (1a)$$

$$L \leq x \leq U \quad (1b)$$

where $0 \leq c < 1$, we will find an ideal non-extended formulation. Please refer to [1] for an explanation on the ideal formulation. In [1] they derived the ideal formulation for a ReLU neuron $y = \max(0, w^T x + b)$, here we extend their approach to leaky ReLU neuron.

2 Big-M formulation

If we denote $m^+ = \sum_i \max(w_i U_i, w_i L_i) + b_i$, $m^- = \sum_i \min(w_i U_i, w_i L_i) + b_i$, then big-M formulation is

$$y \geq c(w^T x + b) \quad (2a)$$

$$y \geq w^T x + b \quad (2b)$$

$$y \leq c(w^T x + b) + (1 - c)m^+ \beta \quad (2c)$$

$$y \leq w^T x + b - (1 - c)m^-(1 - \beta) \quad (2d)$$

$$L \leq x \leq U \quad (2e)$$

$$(2f)$$

This formulation is not ideal, as explained in Fig 1 of [1].

3 Ideal extended formulation

We consider an ideal extended formulation with new slack continuous variables

$$x = x^0 + x^1 \quad (3a)$$

$$y = y^0 + y^1 \quad (3b)$$

$$y^0 = c(w^T x^0 + b(1 - \beta)) \leq 0 \quad (3c)$$

$$y^1 = w^T x^1 + b\beta \geq 0 \quad (3d)$$

$$L(1 - \beta) \leq x^0 \leq U(1 - \beta) \quad (3e)$$

$$L\beta \leq x^1 \leq U\beta \quad (3f)$$

This ideal formulation is extended as it introduces new slack variables x^0, x^1, y^0, y^1 .

4 Ideal non-extended formulation

To derive the ideal non-extended formulation, we want to remove the slack variables x^0, x^1, y^0, y^1 in the ideal extended formulation. We first write x^1, y^0, y^1 as function of x, β, x^0

$$x^1 = x - x^0 \quad (4a)$$

$$y^0 = c(w^T x^0 + b(1 - \beta)) \quad (4b)$$

$$y^1 = w^T x^1 + b\beta = w^T x - w^T x^0 + b\beta \quad (4c)$$

$$y = y^0 + y^1 = w^T x - (1 - c)w^T x^0 + cb + b(1 - c)\beta \quad (4d)$$

Hence we can express $(1 - c)w^T x^0$ as

$$(1 - c)w_1 x_1^0 = w^T x - (1 - c) \sum_{i>1} w_i x_i^0 + cb + b(1 - c)\beta - y \quad (5)$$

We define

$$\bar{L}_i = \begin{cases} L_i & \text{if } w_i > 0 \\ U_i & \text{if } w_i < 0 \end{cases} \quad \bar{U}_i = \begin{cases} U_i & \text{if } w_i > 0 \\ L_i & \text{if } w_i < 0 \end{cases} \quad (6)$$

We have

$$(1 - c)w_1 \bar{L}_1(1 - \beta) \leq (1 - c)w_1 x_1^0 \leq (1 - c)w_1 \bar{U}_1(1 - \beta) \quad (7a)$$

$$(1 - c)w_1(x_1 - \bar{U}_1\beta) \leq (1 - c)w_1 x_1^0 \leq (1 - c)w_1(x_1 - \bar{L}_1\beta) \quad (7b)$$

replacing $(1-c)w_1x_1^0$ with the right-hand side of (16) we get

$$y \leq w^T x + cb + b(1-c)\beta - (1-c)w_1\bar{L}_1(1-\beta) - (1-c) \sum_{i>1} w_i x_i^0 \quad (8a)$$

$$y \leq cw_1x_1 + \sum_{i>1} w_i x_i + cb + b(1-c)\beta + (1-c)w_1\bar{U}_1\beta - (1-c) \sum_{i>1} w_i x_i^0 \quad (8b)$$

$$y \geq w^T x + cb + b(1-c)\beta - (1-c)w_1\bar{U}_1(1-\beta) - (1-c) \sum_{i>1} w_i x_i^0 \quad (8c)$$

$$y \geq cw_1x_1 + \sum_{i>1} w_i x_i + cb + b(1-c)\beta + (1-c)w_1\bar{L}_1\beta - (1-c) \sum_{i>1} w_i x_i^0 \quad (8d)$$

Now we proceed to get rid of x_2^0 , we have

$$\begin{cases} w^T x + cb + b(1-c)\beta - (1-c)w_1\bar{U}_1(1-\beta) - (1-c) \sum_{i>2} w_i x_i^0 - y \\ cw_1x_1 + \sum_{i>1} w_i x_i + cb + b(1-c)\beta + (1-c)\bar{L}_1\beta - (1-c) \sum_{i>2} w_i x_i^0 - y \\ (1-c)w_2\bar{L}_2(1-\beta) \\ (1-c)w_2(x_2 - \bar{U}_2\beta) \end{cases} \leq (1-c)w_w x_2^0 \leq \begin{cases} w^T x + cb + b(1-c)\beta - (1-c)w_1\bar{L}_1(1-\beta) - (1-c) \sum_{i>2} w_i x_i^0 - y \\ cw_1x_1 + \sum_{i>1} w_i x_i + cb + b(1-c)\beta + (1-c)\bar{U}_1\beta - (1-c) \sum_{i>2} w_i x_i^0 - y \\ (1-c)w_2\bar{U}_2(1-\beta) \\ (1-c)w_2(x_2 - \bar{L}_2\beta) \end{cases} \quad (9)$$

So we end up with the constraints

$$y \geq cb + b(1-c)\beta + \sum_{i \in \mathcal{I}} w_i x_i - (1-c)(1-\beta) \sum_{i \in \mathcal{I}} w_i \bar{U}_i + c \sum_{i \notin \mathcal{I}} w_i x_i + (1-c)\beta \sum_{i \notin \mathcal{I}} w_i \bar{L}_i \quad (10a)$$

$$y \leq cb + b(1-c)\beta + \sum_{i \in \mathcal{I}} w_i x_i - (1-c)(1-\beta) \sum_{i \in \mathcal{I}} w_i \bar{L}_i + c \sum_{i \notin \mathcal{I}} w_i x_i + (1-c)\beta \sum_{i \notin \mathcal{I}} w_i \bar{U}_i \quad (10b)$$

where $\mathcal{I}$ is a subset of $\{1, 2, \dots, n\}$. Moreover, (10a) is redundant.

In the end, we obtain the following non-extended ideal formulation

$$y \geq c(w^T x + b) \quad (11a)$$

$$y \geq w^T x + b \quad (11b)$$

$$y \leq cb + b(1-c)\beta + \sum_{i \in \mathcal{I}} (w_i x_i - (1-c)(1-\beta)w_i \bar{L}_i) + \sum_{i \notin \mathcal{I}} (cw_i x_i + (1-c)\beta w_i \bar{U}_i) \quad (11c)$$

$$L \leq x \leq U \quad (11d)$$

Notice that there are exponential number of subsets $\mathcal{I}$. To find $\mathcal{I}$, we consider to take a point $\hat{x}, \hat{y}, \hat{\beta}$ satisfying the constraints in the big-M formulation, with $\hat{\beta}$ being the continuous relaxation $0 \leq \hat{\beta} \leq 1$. Then we take $\mathcal{I}$ as

$$\mathcal{I} = \{i | w_i \hat{x}_i \leq (1 - \hat{\beta})w_i \bar{L}_i + \hat{\beta}w_i \bar{U}_i\} \quad (12)$$

We take an iterative procedure to strengthen the big-M formulation. Say in the current iterations, the strengthened big-M formulation already contains the constraint with $\mathcal{I}_1, \dots, \mathcal{I}_k$, then for an $\hat{x}$ at the boundary of the box $L \leq x \leq U$, we first compute its range of $\hat{\beta}$ under the existing constraint

$$\max(c(w^T \hat{x} + b), w^T \hat{x} + b) \leq cb + b(1 - c)\hat{\beta} + \sum_{i \in \mathcal{I}_k} (w_i x_i - (1 - c)(1 - \hat{\beta})w_i \bar{L}_i) + \sum_{i \notin \mathcal{I}_k} (cw_i x_i + (1 - c)\hat{\beta}w_i \bar{U}_i) \quad (13)$$

Then for each extreme $\hat{x}, \hat{\beta}$, we compute the corresponding index set $\mathcal{I}$ from (12). We then decide whether to enforce this cutting plane, by whether this new cutting plane shrinks the right-hand side of (11c) at $\hat{x}, \hat{\beta}$.

5 Adding bounds on ReLU input

If the lower and upper bound of the ReLU input m^+/m^- is not computed as $\max/\min_{L \leq x \leq U} w^T x + b$ (for example, we compute the bounds by solving an LP/MIP), then the leaky ReLU constraint $y = \max(cw^T x + b, w^T x + b)$ has the following ideal extended formulation

$$x = x^0 + x^1 \quad (14a)$$

$$y = y^0 + y^1 \quad (14b)$$

$$y^0 = c(w^T x^0 + b(1 - \beta)) \leq 0 \quad (14c)$$

$$y^1 = w^T x^1 + b\beta \geq 0 \quad (14d)$$

$$(m^- - b)(1 - \beta) \leq w^T x^0 \leq (m^+ - b)(1 - \beta) \quad (14e)$$

$$(m^- - b)\beta \leq w^T x^1 \leq (m^+ - b)\beta \quad (14f)$$

$$L(1 - \beta) \leq x^0 \leq U(1 - \beta) \quad (14g)$$

$$L\beta \leq x^1 \leq U\beta \quad (14h)$$

Hence we have the condition

$$\begin{aligned}
& \begin{cases} (1-c)(m^- - b)(1-\beta) - (1-c) \sum_{i>1} w_i x_i^0 \\ (1-c)w^T x - (1-c)(m^+ - b)\beta - (1-c) \sum_{i>1} w_i x_i^0 \\ (1-c)w_1 \bar{L}_1(1-\beta) \\ (1-c)w_1(x_1 - \bar{U}_1\beta) \end{cases} \\
& \leq (1-c)w_1 x_1^0 \leq \\
& \begin{cases} (1-c)(m^+ - b)(1-\beta) - (1-c) \sum_{i>1} w_i x_i^0 \\ (1-c)w^T x - (1-c)(m^- - b)\beta - (1-c) \sum_{i>1} w_i x_i^0 \\ (1-c)w_1 \bar{U}_1(1-\beta) \\ (1-c)w_1(x_1 - \bar{L}_1\beta) \end{cases}
\end{aligned} \tag{15}$$

Using the relationship

$$(1-c)w_1 x_1^0 = w^T x - (1-c) \sum_{i>1} w_i x_i^0 + cb + b(1-c)\beta - y \tag{16}$$

we get

$$\begin{aligned}
& \begin{cases} w^T x + cb + b(1-c)\beta - (1-c)w_1 \bar{U}_1(1-\beta) - (1-c) \sum_{i>1} w_i x_i^0 \\ cw_1 x_1 + \sum_{i>1} w_i x_i + cb + b(1-c)\beta + (1-c)w_1 \bar{L}_1\beta - (1-c) \sum_{i>1} w_i x_i^0 \\ w^T x + b - (1-\beta)(1-c)m^+ \\ c(w^T x + b) + m^- \beta(1-c) \end{cases} \\
& \leq y \leq \\
& \begin{cases} w^T x + cb + b(1-c)\beta - (1-c)w_1 \bar{L}_1(1-\beta) - (1-c) \sum_{i>1} w_i x_i^0 \\ cw_1 x_1 + \sum_{i>1} w_i x_i + cb + b(1-c)\beta + (1-c)w_1 \bar{U}_1\beta - (1-c) \sum_{i>1} w_i x_i^0 \\ w^T x + b - (1-\beta)(1-c)m^- \\ c(w^T x + b) + m^+ \beta(1-c) \end{cases}
\end{aligned} \tag{17}$$

6 Appendix

In this appendix, we try to derive the ideal non-extended formulation for ReLU unit (not the leaky ReLU) unit

$$y = \max(0, w^T x + b) \tag{18a}$$

$$L \leq x \leq U \tag{18b}$$

6.1 big-M formulation

We denote $m^+ = \sum_i \max(w_i L_i, w_i U_i)$, $m^- = \sum_i \min(w_i L_i, w_i U_i)$, and get the following big-M formulation

$$y \geq 0 \quad (19a)$$

$$y \geq w^T x + b \quad (19b)$$

$$y \leq m^+ \beta \quad (19c)$$

$$y \leq w^T x + b - m^-(1 - \beta) \quad (19d)$$

$$L \leq x \leq U \quad (19e)$$

6.2 Ideal extended formulation

Using the multiple-choice trick, we get the following ideal formulation with slack variables

$$x = x^0 + x^1 \quad (20a)$$

$$y = y^0 + y^1 \quad (20b)$$

$$y^0 = 0 \geq w^T x^0 + b(1 - \beta) \quad (20c)$$

$$y^1 = w^T x^1 + b\beta \geq 0 \quad (20d)$$

$$L(1 - \beta) \leq x^0 \leq U(1 - \beta) \quad (20e)$$

$$L\beta \leq x^1 \leq U\beta \quad (20f)$$

6.3 Ideal non-extended formulation

To derive the ideal non-extended formulation, we will need to remove the slack variable x^0, x^1, y^0, y^1 from the ideal extended formulation. To do so, we first remove x^1, y^0, y^1 as

$$x^1 = x - x^0 \quad (21)$$

$$y^0 = 0 \quad (22)$$

$$y^1 = w^T(x - x^0) + b\beta = y \quad (23)$$

Hence we have

$$w^T x^0 = w^T x + b\beta - y \quad (24)$$

$$\Leftrightarrow w_1 x_1^0 = w^T x + b\beta - \sum_{i>1} w_i x_i^0 - y \quad (25)$$

Without loss of generality, we assume $w_i > 0 \forall i$, hence

$$w_1 L_1(1 - \beta) \leq w_1 x_1^0 \leq w_1 U_1(1 - \beta) \quad (26a)$$

$$w_1 x_1 - w_1 U_1 \beta \leq w_1 x_1^0 \leq w_1 x_1 - w_1 L_1 \beta \quad (26b)$$

Substituting $w_1x_1^0$ in the inequalities (26) with the right hand side of (25), we get the following inequalities

$$y \geq w^T x + b\beta - \sum_{i>1} w_i x_i^0 - w_1 U_1(1 - \beta) \quad (27a)$$

$$y \geq \sum_{i>1} w_i x_i + b\beta - \sum_{i>1} w_i x_i^0 + w_1 L_1 \beta \quad (27b)$$

$$y \leq w^T x + b\beta - \sum_{i>1} w_i x_i^0 - w_1 L_1(1 - \beta) \quad (27c)$$

$$y \leq \sum_{i>1} w_i x_i + b\beta - \sum_{i>1} w_i x_i^0 + w_1 U_1 \beta \quad (27d)$$

these inequalities remove the slack variable x_1^0 .

Now we proceed to remove the slack variable x_2^0 , we have

$$\begin{aligned} \max \begin{cases} w^T x + b\beta - \sum_{i>2} w_i x_i^0 - y - w_1 U_1(1 - \beta) \\ \sum_{i>1} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 - y + w_1 L_1 \beta \\ w_2 L_2(1 - \beta) \\ w_2 x_2 - w_2 U_2 \beta \end{cases} &\leq w_2 x_2^0 \leq \\ \min \begin{cases} w^T x + b\beta - \sum_{i>2} w_i x_i^0 - y - w_1 L_1(1 - \beta) \\ \sum_{i>1} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 - y + w_1 U_1 \beta \\ w_2 U_2(1 - \beta) \\ w_2 x_2 - w_2 L_2 \beta \end{cases} & \end{aligned} \quad (28)$$

we get the inequalities

$$y \geq w^T x + b\beta - \sum_{i>2} w_i x_i^0 - \sum_{i \leq 2} w_i U_i (1 - \beta) \quad (29a)$$

$$y \geq \sum_{i \neq 2} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 - w_1 U_1 (1 - \beta) + w_2 L_2 \beta \quad (29b)$$

$$y \geq \sum_{i>1} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 + w_1 L_1 \beta - w_2 U_2 (1 - \beta) \quad (29c)$$

$$y \geq \sum_{i>2} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 + \sum_{i \leq 2} w_i L_i \beta \quad (29d)$$

$$y \leq w^T x + b + b\beta - \sum_{i>2} w_i x_i^0 - \sum_{i \leq 2} w_i L_i (1 - \beta) \quad (29e)$$

$$y \leq \sum_{i>1} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 + w_1 U_1 \beta - w_2 L_2 (1 - \beta) \quad (29f)$$

$$y \leq \sum_{i \neq 2} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 - w_1 L_1 (1 - \beta) + w_2 U_2 \beta \quad (29g)$$

$$y \leq \sum_{i>2} w_i x_i + b\beta - \sum_{i>2} w_i x_i^0 + \sum_{i \leq 2} w_i U_i \beta \quad (29h)$$

If we follow this procedure to remove the remaining slack variables, we end up with the inequalities

$$y \geq \sum_{i \in \mathcal{I}} w_i x_i + b\beta - \sum_{i \in \mathcal{I}} w_i U_i (1 - \beta) + \sum_{i \notin \mathcal{I}} w_i L_i \beta \quad (30a)$$

$$y \leq \sum_{i \in \mathcal{I}} w_i x_i + b\beta - \sum_{i \in \mathcal{I}} w_i L_i (1 - \beta) + \sum_{i \notin \mathcal{I}} w_i U_i \beta \quad (30b)$$

where $\mathcal{I}$ is a subset of the set $\{1, 2, \dots, n\}$.

References

- [1] Ross Anderson, Joey Huchette, Christian Tjandraatmadja and Juan Pablo Vielma *Strong mixed-integer programming formulations for trained neural networks*, Mathematical Programming, 2020